MediMatch: Project Defense & Viva Guide

NLP-Driven Medicine Substitute & Symptom Recommender | Everything You Need to Know for Evaluation

1. Executive Summary & Problem Statement

The Problem: In retail pharmacies, high-demand commercial brand-name medications (e.g. **Panadol**, **Brufen**, **Augmentin**) consistently experience supply-chain stockouts, regional shortages, or predatory retail pricing. The average patient leaves without relief simply because they lack pharmacological knowledge that generic counterparts share the **exact same active ingredient**.

The NLP Solution: MediMatch solves this information asymmetry through text processing and information retrieval. It parses complex unstructured composition strings, corrects user misspellings, matches identical active ingredient formulations, recommends clinically similar alternatives using **TF-IDF Cosine Similarity**, and matches colloquial patient symptoms to clinical indications.

2. Dataset & Data Engineering Details

Metric	Value	Significance for Project
Primary Source	1mg Indian Pharma (Kaggle)	11,528 unique medicinal entries across all therapeutic classes
Active Chemical Salts	1,061 unique salts	Extracted and canonicalized into individual active ingredients
Pharmaceutical Brands	770 manufacturers	Allows cross-brand price and availability substitution
Local Augmentation	Pakistani / Global Brands	Augmented 30+ regional staples (Panadol, Risek, Ponstan, etc.)

3. Technical NLP Pipeline (Step-by-Step)

- **Step 1: Text Normalization:** Input strings are lowercased, stripped of non-alphanumeric noise, and token-cleaned. Unstructured runs of clinical text (e.g. *'Pain reliefTreatment of Fever'*) are split using regex lookaheads r'(?<=[a-z\)])(?=[A-Z])'.
- **Step 2: Composition Parsing & Canonicalization:** Formulations like *'Paracetamol (500mg) + Caffeine (65mg)'* are parsed into structured tuples [(salt, strength)]. Regional synonyms are mapped to standard tokens (e.g., *acetaminophen* → *paracetamol*, *amoxicillin* → *amoxycillin*).
- **Step 3: Approximate Fuzzy String Matching:** Misspellings like *'panadoll'* or *'augmntin'* are resolved via **RapidFuzz** utilizing **Levenshtein Distance** and token-weighted ratios (WRatio). Matches score from 0 to 100, activating spelling correction when score ≥ 75.
- **Step 4: Vector Space TF-IDF Representation:** Two distinct vectorizers: (a) **Salt Model:** Custom token analyzer preserving multi-word chemical entities; (b) **Uses Model:** Word unigrams and bigrams (ngram_range=(1, 2)) with sublinear scaling (1 + log(tf)) on clinical indications.

- **Step 5: Hybrid Cosine Similarity Metric:** Cosine similarity computes angular proximity between document vectors: **$\cos(\theta) = (u \cdot v) / (||u|| \, ||v||)$**. Alternative ranking combines **65% chemical salt similarity** + **35% clinical indication similarity**.
- **Step 6: Query Expansion for Lay Symptoms:** Resolves the classic **Vocabulary Mismatch Problem**. A layperson writes *'loose motion'* or *'acidity'*; our query expansion layer maps these into clinical target tokens (*diarrhea*, *gastroesophageal reflux disease*) before TF-IDF scoring.

4. Core Mathematical Formulations

Concept	Formula	Role in MediMatch
Levenshtein Distance	`lev(a,b) = min edits (insert, delete, replace)`	Calculates typo edit distance for fuzzy brand name search
Term Frequency (TF)	`TF(t,d) = f(t,d) / Σ f(t',d)`	Measures occurrence density of medical keywords in 'Uses'
Inverse Document Freq	`IDF(t) = log((1+N)/(1+DF(t))) + 1`	Downweights universal terms ('treatment', 'pain') and boosts distinctive disease terms
Cosine Similarity	`cos(θ) = (u · v) / (||u|| ||v||)`	Measures angle between vectors in 0-to-1 range regardless of text length
Hybrid Alternative Score	`0.65 · Sim(Salts) + 0.35 · Sim(Uses)`	Prioritizes chemical composition first, clinical purpose second

5. Top 6 Teacher / Viva Defense Questions & Winning Answers

Read these carefully. These are the exact conceptual and architectural angles professors inspect:

Q1: Why use TF-IDF rather than a deep learning transformer model like BERT?

Answer: Three crucial reasons: (1) *Explainability & Safety:* In pharmaceutical recommendation, every similarity weight can be audited down to exact n-gram token overlap. Deep models operate as black boxes that can hallucinate false equivalences. (2) *Zero Inference Overhead:* TF-IDF matrices run on CPU in 1-2 milliseconds with no GPU dependency. (3) *Sparse Chemical Precision:* TF-IDF penalizes mismatching salt tokens sharply, whereas semantic embeddings can mistakenly consider two different antibiotics 'close' simply because they appear in similar clinical contexts.

Q2: How do you differentiate between an 'Exact Substitute' and an 'Alternative'?

Answer: An **Exact Substitute** is a bio-equivalent generic drug: it contains the exact same set of active chemical salts (e.g. pure Paracetamol 500mg). An **Alternative** is therapeutic: it treats a similar medical complaint but may have a different chemical base or a combination drug (e.g. Ibuprofen instead of Paracetamol, or Paracetamol + Caffeine). Exact matches use set equivalence; alternatives use TF-IDF cosine similarity.

Q3: How does your system correct typos when a user searches 'panadoll'?

Answer: We implement RapidFuzz WRatio (Weighted Ratio), which calculates the Levenshtein Distance between the user query and our dictionary of known brand stems and salts. When an exact match is missing, the algorithm scans for candidates with similarity ≥ 75%, automatically rerouting to 'Panadol' and informing the user of the correction.

Q4: What is the 'Vocabulary Mismatch Problem' and how did you solve it?

Answer: Patients describe illnesses in everyday conversational language (e.g., 'loose motions', 'stomach burns', 'body pain'), whereas pharmaceutical databases document formal medical conditions (e.g., 'diarrhea', 'gastroesophageal reflux', 'myalgia'). We implemented a **Query Expansion Dictionary** that injects canonical medical terms into user input prior to TF-IDF vectorization.

Q5: How did you adapt an Indian dataset for a local Pakistani user base?

Answer: The 1mg dataset lacks prominent Pakistani brand names like Panadol, Brufen, and Risek. We engineered an augmentation layer in local_brands.json. The system automatically extracts their chemical formulas, matches them with corresponding records in the master database, and inherits the clinical uses and side effects for comprehensive searchability.

Q6: What are the ethical and safety limitations of this project?

Answer: While the system accurately matches chemical ingredients, it does not evaluate patient-specific factors such as allergies, kidney/liver impairment, pregnancy contraindications, or drug-drug interactions. We clearly include safety disclaimers emphasizing that generic substitution should always be verified with a registered pharmacist or physician.